# Contrastive learning for continuous touch-based authentication

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## ABSTRACT

The frequent handling of sensitive information on mobile devices has created an urgent need for robust security measures. Touchscreens, as the primary medium for human-computer interaction, offer an ideal solution for non-intrusive security through continuous authentication based on touch behavior. To address the limitations of existing single-modal binary classification methods, this paper proposes a unified contrastive learning framework. The framework utilizes a Temporal Masked Autoencoder to extract temporal features from multi-sensor data streams. This framework further integrates a Siamese Temporal Attention Convolutional Network and attention mechanisms to jointly model sequential and cross-modal features. Experiments conducted across multiple datasets demonstrate that this method outperforms current mainstream solutions, thus offering a reliable and efficient solution for mobile user authentication.

**Keywords:** Contrastive Learning, Continuous Authentication, Masked Autoencoder, Siamese Network, Temporal Convolutional Networks, Transformers

## 1. INTRODUCTION

The ubiquity of smartphones in managing sensitive data and financial transactions has heightened the demand for robust security. While conventional methods like PINs and static biometrics are common, they remain vulnerable to spoofing[2]. Behavioral biometrics, particularly touch dynamics, offer a promising solution for continuous authentication due to their unobtrusiveness and stability in various physical environments, in contrast to gait or motion-based sensors.

Existing touch-based systems often rely on handcrafted features or traditional classifiers, which fail to capture the complex temporal dependencies inherent in user interactions. This limitation reduces their discriminative power and robustness in real-world scenarios.

To bridge this gap, we propose TouchSeqNet, a Siamese framework integrated with a Temporal-Attentive Convolutional Network. The model utilizes a Temporal Masked Autoencoder (TMAE) for self-supervised pretraining to learn robust temporal representations. These are then refined within a Siamese architecture featuring a finger channel attention mechanism that adaptively emphasizes discriminative features across sequences for precise identity verification.

We evaluate TouchSeqNet on three public datasets—BioIdent, Touchalytics, and BehavePassDB—which together cover a wide range of devices, gestures, and usage scenarios. This paper makes the following key contributions: (1) We propose TouchSeqNet, a novel Siamese architecture that integrates a pretrained TMAE with the TouchSeq framework, to capture both local and long-range temporal dependencies in user behavior, (2) Extensive experiments on three datasets demonstrate that TouchSeqNet outperforms existing methods in accuracy, F1 score, and AUC.

## 2. RELATED WORK

A wide range of behavioral biometric modalities have been explored for continuous user authentication on mobile devices, including touch dynamics[10,11], motion sensor signals[2,5], keystroke dynamics[1], and gait patterns[10]. These studies demonstrated the effectiveness of mobile behavioral authentication in both constrained and unconstrained environments.

Among these modalities, touch-based biometrics have drawn significant attention due to their unobtrusiveness and stability. Frank et al[11]. pioneered large-scale evaluation of touch features, achieving 2% EER with multi-gesture fusion. More recent works leveraged CNNs and multimodal representations to improve classification accuracy [22].

Motion and sensor-based methods also show promising performance, especially when combining CNNs or LSTMs with accelerometer and gyroscope data[2,5]. Similarly, keystroke dynamics have been modeled with RNNs[1] and Transformer hybrids, while gait recognition systems have adopted metric learning and deep CNNs for robust identification.

Despite encouraging progress, many existing methods rely on static or handcrafted features and shallow models, limiting their ability to capture the sequential and dynamic nature of user behavior. Moreover, few approaches incorporate self-supervised learning or pretraining strategies, which are crucial for generalization in real-world deployment.

## 3. METHODOLOGY

### 3.1. TMAE Pre-trained Model Architecture

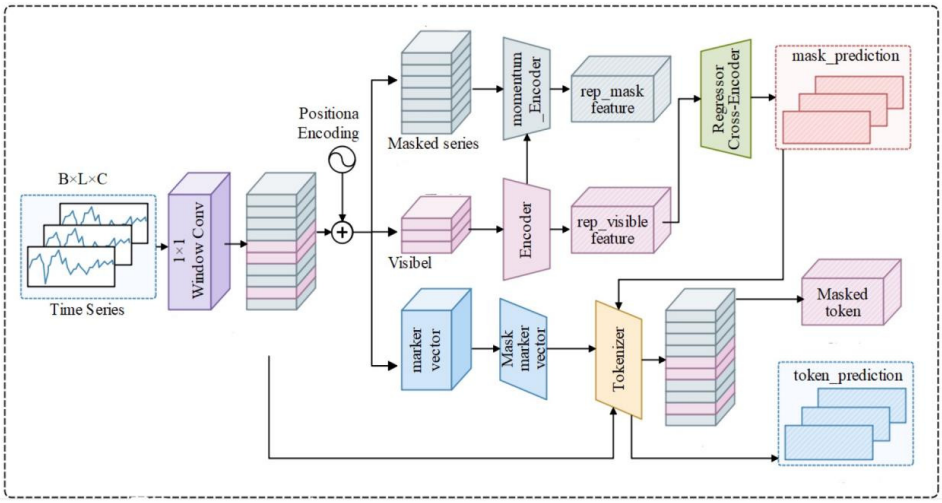


Figure 1. An illustration of the pre-training process, where a time series input is masked and reconstructed using a transformer-based encoder-decoder architecture.

As illustrated in Figure 1, the model first extracts local time series features using a window-based convolutional projection. It then splits the features into two branches: one generates discrete token embeddings, while the other encodes visible features with a Transformer encoder and generates target representations for masked features using a momentum encoder. Finally, a cross-attention regressor predicts masked representations and a decoder reconstructs discrete codewords[14].

To enhance the information density of basic semantic units, we first apply a window slicing strategy to divide the continuous touch sequence into non-overlapping, fixed-length sub-series[15]. Formally, let the original touch sequence be $X \in \mathbb{R}^{T\times C}$. where T denotes the sequence length and C the number of channels. We slice X into non-overlapping windows of size W, with each window corresponding to a sub-series of length W, and perform a 1D convolutional projection on each window to convert the original sequence into a new embedded sequence $Z \in \mathbb{R}^{d\times m}$,where $d=\lfloor T/W \rfloor$ is the number of windows and m the model embedding dimension.

This strategy reduces temporal redundancy while ensuring that each sub-series contains enough semantic information, further enhancing the self-supervised learning process[7].

We convert each sub-series window Z into discrete embeddings using a Tokenizer, which maps raw segments into a compact latent space. Unlike handcrafted features, the Tokenizer supports end-to-end learning for effective representation of touch dynamics.

The Tokenizer module transforms each input sub-series $s_{i:j} \in \mathbb{R}^{\sigma\times m}$ into a continuous embedding representation $E \in \mathbb{R}^{\sigma\times m}$. Each embedding $E_i \in \mathbb{R}^m$ is then projected into a vocabulary space of size K using a linear layer:

$$p_i = \mathrm{softmax}(WE_i + b), i \in \sigma \tag{1}$$

where $W \in \mathbb{R}^{m\times k}$ and $b \in \mathbb{R}^k$ are learnable parameters. This projection generates a probability distribution over the codebook entries[12].

To enable end-to-end differentiability, we adopt Gumbel-Softmax trick to approximate discrete sampling during training. The final discrete token $T_i$ for each position is selected via:

$$T_i = \arg^{\max}{}_j p_i^{(j)}, j \in K \tag{2}$$

Through this process, the windowed feature sequence Z is mapped into a discrete token sequence $T = \{T, T_2, \cdots, T_d\}$, where each token encodes a local temporal pattern. These discrete representations provide a compact and informative abstraction of the raw input, facilitating downstream tasks such as classification and anomaly detection.

The TMAE model's self-supervised masking strategy reconstructs hidden representations of masked windows and predicts their discrete tokens, allowing it to learn temporally structured semantic features from touch dynamic data. To preserve positional information after windowing, positional encoding is added to the sequence embedding Z to produce $Z^p$. The $Z^p$ sequence is then randomly split into visible tokens $Z^v$ and masked tokens $Z^m$, with the discrete token sequ-ence T divided consistently into $T^v$ and $T^m$.

We establish two core pretext tasks (masked representation regression and discrete codeword prediction) and employ a momentum-based encoding strategy and update mechanism. We adopt a dual-encoder structure comprising a primary encoder and a decoupled encoder, which share an identical Transformer architecture yet adopt distinct update strategies. The primary encoder encodes visible inputs to generate the visible representation $R_v$, while the gradient-free update decoupled encoder uses a momentum update strategy to encode the mask markers to generate the target representation $R_m$. A cross-attention-based regressor then leverages the contextual information of visible tokens to predict the masked representation $\vec{R}_m$.

To enhance semantic learning, we introduce a discrete codeword prediction task, where the tokenizer discretizes latent features via Gumbel-Softmax, and the model infers the codeword distribution $P\left(\vec{T}_m\right)$ from reconstructed embeddings against the ground-truth $T_m$.

The momentum encoder parameters are updated by the exponential moving average of the primary encoder weights[6]:

$$\theta_m = \mu\theta_m + (1-\mu)\theta_e \tag{3}$$

where $\theta_m$ and $\theta_e$ denote the parameters of the momentum and primary encoders.

Our model's self-supervised objective includes two loss components: the alignment loss $\mathcal{L}_{\text{align}}$(MSE), between the momentum encoder's target representations and the regressor's predictions) and the prediction loss $\mathcal{L}_{\text{pred}}$(MSE) cross-entropy for the predicted token distribution vs. ground-truth discrete tokens). Auxiliary metrics (Hits, NDCG@10) are also tracked for evaluation. The final loss is a weighted sum of the two terms:

$$\mathcal{L}_{\text{loss}} = \alpha\mathcal{L}_{\text{align}} + \beta\mathcal{L}_{\text{pred}} \tag{4}$$

where α and β are hyperparameters controlling the contribution of each loss component.

### 3.2. TouchSeqNet Architecture

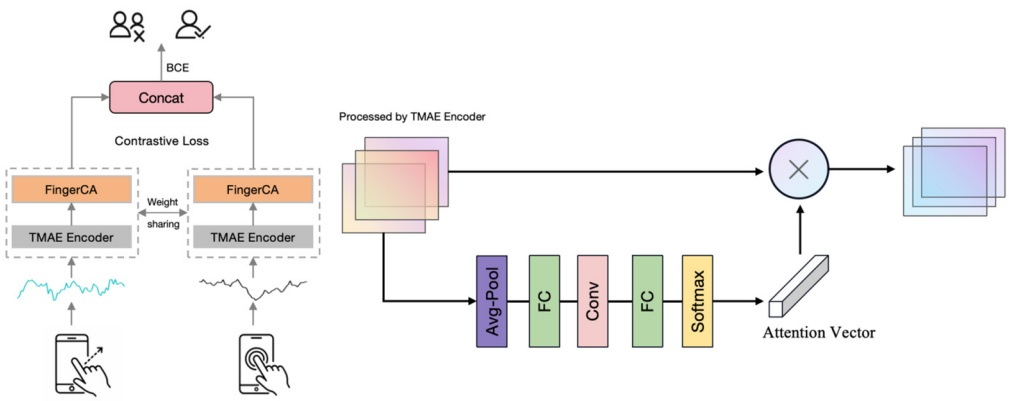


Figure 2. Mobile device user identity authentication model architecture TouchSeqNet and FingerCA.

As shown in Figure 2, we propose TouchSeqNet, a contrastive learning framework for continuous authentication based on dynamic touch. This framework transfers the pre-trained temporal encoder of TMAE, employs window-based convolutional projection and a multi-layer self-supervised Transformer to extract local-contextual temporal features from touch sequences, and obtains highly discriminative user representations via contrastive learning.

This transfer learning initializes the model with touch dynamics-aligned representations and behavioral biometrics-tailored inductive bias, enabling subsequent layers to refine identity-specific features. As shown in Figure 2, To enhance discriminability after temporal modeling, we incorporate the FingerCA channel attention module. For each input pair, we obtain perform temporal average pooling on both, and concatenate the results to form $z = \{z_1|z_2\} \in \mathbb{R}^{2m}$, A fully connected classification head predicts the same-user probability for the pair, with the binary classification trained end-to-end via a hybrid loss. To optimize both representation learning and classification accuracy together, we use a hybrid loss that combines contrastive loss and bce loss. For embeddings $z_1$ and $z_2$ with label $y \in 0,1$(1 for the same user, 0 for different users):

$$\mathcal{L}_{\text{bce}} = -y\log(\hat{y}) - (1-y)\log(1-\hat{y}) \tag{5}$$

The final training objective is a weighted sum of both terms:

$$\mathcal{L} = \lambda_1 \mathcal{L}\text{contrastive} + \lambda_2 \mathcal{L}_{\text{bce}} \tag{6}$$

where $\lambda_1$ and $\lambda_2$ are hyperparameters balancing the two losses.

## 4. EXPERIMENTS

In this section, we describe our experimental protocol for evaluating TouchSeqNet. The evaluation consists of two stages: self-supervised pretraining of the TMAE encoder on unlabeled touch dynamics data, and fine-tuning the TouchSeqNet on labeled data. We conduct experiments on three public datasets: BehavePassDB[8], Touchalytics[11] and BioIdent[3].

### 4.1. Experimental Setup

Pretraining uses a 256 hidden dimension, Adam optimizer with fixed 0.01 learning rate, and a uniform batch size of 128; all datasets follow a 60%:20%:20% train-validation-test split. The TMAE Transformer encoder has 8 layers (4 attention heads, two-layer feed-forward each) with a global 0.2 dropout rate, plus a 4-layer decoupled encoder for masked position contextual representation extraction. For each dataset, 80% of users and their session data are for pretraining and 20% for fine-tuning; window size is 8, masking ratio 40%, tokenizer codebook vocabulary size 192, and all shared pretraining hyperparameters are retained for fine-tuning. The default touch sequence length is set to 128.

Prior to the windowing operation, we perform data preprocessing to eliminate noise and outliers from the raw sensor data. Specifically, we adopt the Winsorization method to cap the extreme values by truncating the top and bottom 5% of the data distribution.

Table 1. Performance comparison of all models on Ffinger, BioIdent, and Touchalytics datasets (Accuracy / F1 / AUC).

| Methods | BehavePassDB | | | Biodent | | | Touchalytics | | |
|---|---|---|---|---|---|---|---|---|---|
| | Acc | F1 | AUC | Acc | F1 | AUC | Acc | F1 | AUC |
| TCN[17] | 0.9558 | 0.9582 | 0.9656 | 0.9052 | 0.9088 | 0.9050 | 0.9001 | 0.9033 | 0.9007 |
| Gate-transformer[13] | 0.9343 | 0.9384 | 0.9332 | 0.9794 | 0.9789 | 0.9792 | 0.9852 | 0.9852 | 0.9852 |
| LSTM[4] | 0.8851 | 0.8820 | 0.8851 | 0.9656 | 00.9638 | 0.9647 | 0.9713 | 0.9715 | 0.9713 |
| InceptionTime[16] | 0.8447 | 0.8644 | 0.8402 | 0.9886 | 0.9878 | 0.9885 | 0.9755 | 0.9754 | 0.9756 |
| TSLANET[9] | 0.7247 | 0.7453 | 0.7227 | 0.9831 | 0.9833 | 0.9831 | 0.9415 | 0.9426 | 0.9419 |
| ours | 0.9769 | 0.9770 | 0.9769 | 0.9902 | 0.9908 | 0.9907 | 0.9908 | 0.9907 | 0.9908 |

### 4.2. Performance Evaluation

Self-supervised learning excels at learning transferable representations from unlabeled data through pretraining and fine-tuning. However, touch dynamics datasets vary significantly in their acquisition and design. We thus perform independent self-supervised pretraining on Touchalytics, BioIdent and BehavePassDB, yielding representations tailored to each dataset for robust downstream fine-tuning initialization.

As shown in Table1, the proposed TouchSeqNet consistently outperforms all baseline methods across the three evaluated datasets. On the Touchalytics dataset, it achieves the highest performance across all metrics, demonstrating strong robustness in modeling fine-grained, user-specific interaction patterns. In comparison, models such as TSLANet and LSTM show significant performance drops, indicating their limitations in capturing such behavioral variability.

On the public benchmarks BioIdent and BehavePassDB, TouchSeqNet also reaches near-perfect results in terms of accuracy, F1 score, and AUC, outperforming or matching all competing approaches. These results highlight the model's superior generalization and discriminative capabilities, validating its effectiveness for continuous touch-based user authentication in practical scenarios.

### 4.3. Ablation Study

To verify the effectiveness of each component, we conduct a step-by-step ablation study, as reported in Table 2.

Table 2. Ablation study on the effectiveness of core components(AUC%).

| Pre-training | FingerCA | BehavePassDB | Biodent | Touchalytics |
|---|---|---|---|---|
| × | × | 83.31 | 93.99 | 98.06 |
| ✓ | × | 93.23 | 98.34 | 89.68 |
| × | ✓ | 88.47 | 87.11 | 92.01 |
| ✓ | ✓ | 97.69 | 99.07 | 99.08 |

Removing the pretrained encoder results in a significant AUC drop across all datasets. On Touchalytics, for instance, the AUC decreases from 99.08% to 92.01%, which directly validates the effectiveness of transfer learning. The pretrained encoder learns generalized touch dynamics features from large-scale, unlabeled data, thereby providing robust, initialized representations for downstream tasks. When relying solely on the pretrained encoder, the model achieves the lowest overall scores on all datasets. On Touchalytics, the AUC is only 89.68%, confirming that pretraining alone is insufficient and must be complemented by downstream temporal modeling to fully exploit identity-related discriminative information. Additionally, ablation of the channel attention mechanism FingerCA also results in a significant AUC decrease across datasets. On Touchalytics, the AUC drops from 99.08% to 89.68%, demonstrating its ability to accurately capture inter-channel feature correlations, weight critical information, and effectively enhance feature discriminability. These results highlight the complementary strengths of the pretrained encoder and the channel attention mechanism.

Additionally, we further validated the model performance with hidden dimensions of 128 and 512 on the Touchalytics dataset, achieving accuracies of 96.68% and 97.12%, respectively.

## 5. CONCLUSION

We presented TouchSeqNet, a contrastive learning framework designed for continuous user authentication via touch dynamics. By integrating self-supervised pretraining, hierarchical temporal modeling, and attention mechanisms, TouchSeqNet learns rich, discriminative representations without requiring handcrafted features or domain-specific heuristics.

Our architecture demonstrates the synergistic benefits of transfer learning and structured temporal modeling. The pretrained encoder offers a generalizable feature space, while the FingerCA block enhances temporal resolution and discriminability—together enabling robust identity verification across varying users and contexts.

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